

Thanh-Vinh Tong

+84-363-955-684 | vincent.tong369@gmail.com |  [Github](#)

Perth, Australia (Open to relocating to Vietnam)

PROFESSIONAL SUMMARY

Backend-focused Software Engineer with 2+ years of experience building backend services, data pipelines, and end-to-end web applications.

EXPERIENCE

- **Rate Pulse**  June 2025 - Present
Software Engineer Australia
 - Built an end-to-end FX intelligence platform combining exchange-rate tracking, tax- and fee-inclusive conversion, historical trends, market news, and user personalisation in a single workflow, helping users compare market information and make faster decisions.
 - Developed backend **microservices** in **Go**, using **REST** and **gRPC** for client-server communication. Contributed to building and maintaining **PostgreSQL** databases. Used **Redis** to cache **frequently accessed, rarely changing data**, reducing **P95 latency** at **20 QPS** from **7–10 seconds** to approximately **250 ms**.
 - Developed a multi-source scraping engine to collect and normalise structured market data using **Python**, **Selenium**, and **GitHub Actions**.
 - Built **CI/CD** pipelines with **GitHub Actions**. Migrated services from **AWS** to a lower-cost cloud deployment on **DigitalOcean**, using **Docker** and **Kubernetes**, reducing infrastructure costs.
 - Built and deployed a responsive frontend using **Next.js**, and **TypeScript** on **Vercel**, with server-side rendering, parallel data fetching across backend services, and typed API responses to reduce integration errors.
- **Lions Eye Institute**  January 2025 – May 2025
Software Engineer Intern Australia
 - Developed an internal web application using **Python**, and **FastAPI**. Leveraged **embedded SQLite database**, to manage data efficiently, **minimize memory usage**, and **database network I/O overhead**, on a **low-resource server**. Reduced the review process from around **1 hour** to **approximately 10 minutes**.
 - Developed an internal web using **Next.js**, **ASP.NET Core**, and **PostgreSQL** for managing laboratory workflows.
- **Samsung**  August 2023 – October 2023
Software Engineer Intern Vietnam
 - Developed an end-to-end desktop application using **.NET** and a **SQL Server** database to manage projects.
 - Deployed an AI model for display-panel defect detection, reducing manual inspection time by **40%**.
- **VNPT Group**  June 2022 – June 2023
Data Engineer Vietnam
 - Developed, and migrated, multi big data pipelines, processing approximately **100 millions** data records daily, from using **Pandas**, to **Spark**, and **Hadoop**. Reduced data pipelines runtime by **3 times** better.
 - Developed backend services for wrapping core **AI models**, using **Python**, **FastAPI**, and **PyTorch**.
- **Samsung**  Aug 2022 – Sep 2022
Software Engineer Intern Vietnam
 - Developed *Just Jump*, an offline casual Android game inspired by *Bounce*, using **Kotlin**; designed short, repeatable gameplay for mobile users and separated player movement, obstacles, scoring, and game-state logic to simplify future updates.

CERTIFICATIONS

- **AWS Certified Developer – Associate** February 2026 – February 2029
Amazon Web Services 
 - Demonstrated skills to develop, deploy, and debug cloud-based applications using AWS best practices.

AWARDS

- **Out of the Box Idea Award** March 2025
VAIT Hackathon 2025 
 - Recognised for presenting a creative and unconventional solution at the VAIT Hackathon.
- **Best Trainee Award** October 2023
Samsung Display Vietnam 
 - Received for overall performance and project contributions during the software engineering internship.
- **Second place** September 2022
Samsung Application Development Contest - 2022
 - Placed second for a creative casual-game concept designed to appeal to mobile users.

EDUCATION

• The University of Western Australia <i>Master of Information Technology, Software Systems Specialisation</i>	<i>February 2024 – December 2025</i> Perth, Australia
• Hanoi University of Science and Technology <i>Bachelor of Information Technology</i>	<i>August 2019 – September 2023</i> Hanoi, Vietnam
• HUS High School for Gifted Students <i>Mathematics Specialisation</i>	<i>August 2016 – July 2019</i> Hanoi, Vietnam

PROJECTS

• Book Borrow <i>FastAPI, Next.js, AWS RDS, Stripe, Google OAuth</i>	<i>August 2025 – November 2025</i> [ ]
◦ Built an end-to-end peer-to-peer book marketplace for a Western Australian non-profit organisation.	
◦ Developed backend services, using Python , and FastAPI , with application data stored on AWS RDS , and built the user-facing application using Next.js .	
◦ Integrated Google OAuth for authentication and Stripe for payment processing, supporting account, book-listing, borrowing, and transaction workflows.	

SKILLS

- **Programming Languages:** Go, Python, C#, TypeScript/JavaScript, SQL
- **Backend & Web Technologies:** Gin, FastAPI, ASP.NET Core, REST APIs, gRPC, Next.js, React
- **Databases & Data Processing:** PostgreSQL, Redis, MongoDB, SQLite, PySpark, Hadoop
- **Cloud & DevOps:** AWS, Docker, Kubernetes, GitHub Actions, Git, Linux